Fig. 1 U-Chrom at a glance: many assays → one data model → analysis, a linked browser and an open atlas

a Many assays, three kinds of data

each with its own formats and viewers

glyph

Single-cell Hi-C (+ RNA, ATAC, 3-D)

HiRES · Dip-C · GAGE-seq · Droplet / Paired Hi-C
dscHi-C · scHiCAR · Uni-C

contact map per cell

.pairs · .scool | Juicebox, HiGlass

glyph

Spatial Hi-C (+ RNA, ATAC)

Hi-C-RNA · spatial Hi-C · spatial ATAC-Hi-C

contact map per spot, tissue position

pairs + Seurat | no viewer for per-spot maps

glyph

Chromatin tracing (+ IF, RNA)

DNA seqFISH+ · ORCA · Hi-M · MERFISH

x, y, z per traced locus

FOF-CT, CSV | custom scripts

b ChromData: one index for all three

cells (rows) × genomic bins (columns); spots = cell › trace › bin with x, y, z;
contact maps linked per cell, not copied; signals per spot, per bin, per cell

the cells × bins schematic
(current Fig. 1a, refined)

c One store

x.chromdata.zarr
Zarr + Parquet, open spec

read in parts:
one cell, one chromosome

from disk or
over HTTP (S3, R2)

maps linked
or embedded

.cdz: one zip file

d Analysis in Python

one API on every kind of data

subset / stream
cell, chromosome, trace;
larger than memory

aggregate
pseudo-bulk maps, median
distance maps per group

convert
3-D structures ↔ contacts

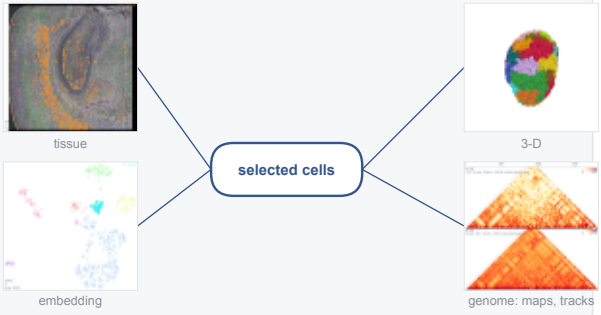
call
compartments, TADs, loops

features
per locus, per cell

link
.scool, .h5ad, SpatialData

e One browser: views linked through cells

select cells anywhere; every view follows (people and agents: the same actions)



f Where you work

notebook
Python API, Jupyter

your laptop
python -m uchrom.browser

the web
hosted, no install

coding agents
WebMCP actions, skills

g An open atlas, ready to browse

22 datasets · 11 studies · 315,516 cells and spots · 2.0 × 10[2] 3-D loci · 25.6 GB · contacts per cell checked against the published numbers

spatial Hi-C (10)

single-cell Hi-C (11)

tracing (1)

